REVA UNIVERSITY
Bengaluru, India
Established as per the section 2(f) of the UGC Act, 1956,
Approved by AICTE, New Delhi

School of Computing Science and Engineering

Course Code: B22EHS626	AI APPLICATIONS DEVELOPMENT	**Year: 2025-2026**
		Sem: 6th

Project Group Members Details

Group No: B10	**Group Name: AIDS – B10**					
Sl. No	**SRN**	**Full Name**	**Sec**	**Mobile No.**	**Email_ID**	**Sign**
1	**R23EK026**	**NISHA YADAV**	**AI&DS-B**	**8971682389**	**yadav16ni@gmail.com**	
2	**R23EH080**	**NAMEERA SANIYA**	**AI&DS-B**	**8073531848**	**ugcet2300579@reva.edu.in**	
3	**R23EH109**	**RUHI SHARMA**	**AI&DS-B**	**7676678051**	**23020301429@reva.edu.in**	
4	**R23EH072**	**MEGHANA U**	**AI&DS-B**	**8088980245**	**meghss1236@gmail.com**	
5	**R23EO033**	**SANJANA R**	**AI&DS-B**	**8431672247**	**sanjanaprasad2903@gmail.com**	
6	**R23EH111**	**SAGARIKA DS**	**AI&DS-B**	**8951197503**	**dssagarika012@gmail.com**	

Project Details

Project Title:	**AgroLedger** Farmer Provenance System for Agricultural Supply Chain	
Guide Details:	**Name: Dr. RAMARAJ SRINIVASAN**	**Mobile No: 9845199854**
Remarks by Guide:		**Guide Signature** **Date:**

Tables of content

SL No.	Content	Page No.
1	Abstract	3
2	Introduction	4
3	Literature Survey	5
4	Objectives	6
5	Methodology	6-7
6	Modules Identified	8
7	Work Progress / Plan & Implementation	9
8	Conclusions	10
9	References	11

Abstract:

AgroLedger is a QR-based agricultural traceability system designed to improve transparency, trust, and accountability in the agricultural supply chain. In traditional systems, buyers depend on verbal claims regarding product quality, origin, and handling, which often leads to misinformation and reduced trust. The proposed system allows farmers to digitally register crop batch details, including farmer identity, crop type, location, harvest date, and quantity. Each batch is assigned a unique Batch ID and linked to a QR code, enabling easy access to product information.

AgroLedger implements a ***hybrid storage model****, where data is stored on the blockchain when a wallet is connected, ensuring immutability and security. In cases where blockchain access is unavailable, the system uses cloud storage (JSONBin) as a fallback to maintain accessibility and usability.*

Buyers can scan the QR code to instantly verify product details through a smart search mechanism that prioritizes blockchain verification and falls back to cloud storage when required. The system clearly indicates verification status, improving trust between farmers and buyers.

AgroLedger is designed to be simple, scalable, and cost-effective, making it suitable for small-scale farmers and real-world deployment.

I. Introduction:

Agriculture plays a vital role in the economy and food supply chain, yet it faces significant challenges related to transparency and trust. In traditional agricultural systems, buyers and consumers often have limited access to reliable information about the origin, quality, and handling of produce. This lack of traceability creates uncertainty and reduces confidence in product claims.

Small-scale farmers are particularly affected, as they depend on intermediaries to sell their produce. These intermediaries often control pricing and communication, making it difficult for farmers to prove the authenticity and quality of their products. As a result, farmers may not receive fair value for their produce, while buyers struggle to differentiate between genuine and misleading claims.

With advancements in technology, there is a growing need for digital solutions that can improve transparency in agricultural supply chains. Technologies such as blockchain and QR codes offer effective ways to store and access product information securely. Blockchain ensures that data is tamper-proof and trustworthy, while QR codes enable quick and easy access to information through simple scanning.

AgroLedger is developed as a farmer-centric provenance system that addresses these challenges by enabling digital traceability of agricultural produce. The system allows farmers to register crop details such as crop type, location, harvest date, and quantity. Each batch is assigned a unique ID and linked to a QR code, which serves as a digital identity for the product.

When a buyer scans the QR code, they can instantly view detailed information about the product, ensuring transparency and trust. The system uses a hybrid storage approach, where data is stored on the blockchain when available and on cloud storage as a fallback, ensuring both reliability and accessibility.

The platform is designed to be simple and user-friendly, making it accessible even to farmers with limited technical knowledge. By reducing dependency on intermediaries and enabling direct verification, AgroLedger helps build trust between farmers and buyers.

In conclusion, AgroLedger provides a practical solution for improving transparency in agricultural supply chains by leveraging modern technologies, ultimately promoting fair trade and informed decision-making.

II. Literature Survey:

1. "Blockchain-Based Food Supply Chain Traceability System" – Tian (2017) & "Blockchain for Sustainable Supply Chain Management" – Saberi et al. (2019)

These studies highlight the importance of blockchain technology in improving transparency, traceability, and trust in supply chains. Blockchain provides a decentralized and immutable ledger, ensuring that once data is recorded, it cannot be altered. This helps in preventing fraud and maintaining accountability across all stages of the supply chain.

The research demonstrates that blockchain can effectively track agricultural products from farm to consumer, ensuring authenticity and reliability of data. However, these systems often require high computational resources, technical expertise, and infrastructure, making them difficult to implement for small-scale farmers.

AgroLedger adopts the core advantage of blockchain—secure and tamper-proof data storage—while overcoming its limitations by introducing a simplified interface and a fallback cloud storage mechanism.

2. "A Survey on Food Traceability Systems" – Bosona and Gebresenbet (2013) & "QR Code Based Product Information System" – Law and So (2010)

These papers focus on the role of traceability systems and quick-access technologies in improving supply chain visibility. Traditional traceability systems are often manual and lack real-time data access, leading to inefficiencies and lack of trust.

The use of QR codes has been identified as an effective solution for providing instant access to product information through mobile devices. QR codes are cost-effective, easy to implement, and widely accessible, making them suitable for large-scale adoption.

AgroLedger combines these concepts by using QR codes to provide instant access to agricultural product data, enabling buyers to verify product details quickly and efficiently.

III. Objectives:

The primary objectives of this project are:

- To design a digital platform for agricultural product traceability
- To generate unique batch IDs and QR codes for each crop batch
- To enable quick and reliable product verification
- To implement hybrid storage combining blockchain and cloud
- To ensure cross-device accessibility and data synchronization
- To reduce dependency on intermediaries
- To improve trust and transparency in agricultural transactions

IV. Methodology:

The AgroLedger system follows a structured workflow to ensure efficient data creation, storage, and verification in the agricultural supply chain.

1. Data Collection

Farmers enter crop-related details such as farmer name, crop type, location, quantity, harvest date, and additional notes using a simple user interface. This ensures that all essential information about the product is recorded digitally.

2. Data Processing and Batch Creation

Once the farmer submits the form, the system validates the data and generates a unique Batch ID (e.g., FT-XXXXX). This ID acts as a unique identifier for each crop batch.

3. Storage Decision Logic

The system checks whether a blockchain wallet is connected:

- If connected, the data is stored securely on the blockchain.
- If not connected, the data is stored in cloud storage (JSONBin).

This ensures flexibility and continuous operation.

4. Data Storage

- Blockchain Storage**:** Provides secure, tamper-proof, and immutable data storage. A transaction hash is generated for verification.
- Cloud Storage: Provides fast access and cross-device synchronization.

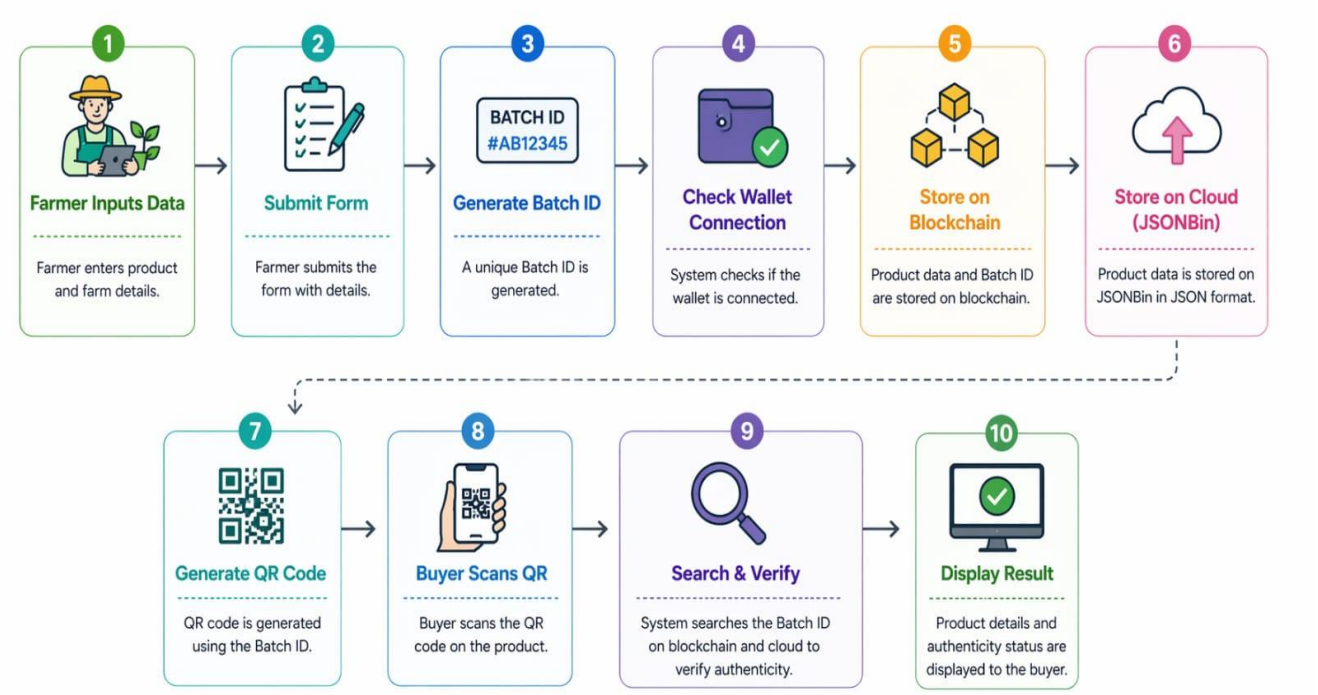
5. QR Code Generation
A unique verification URL is generated using the Batch ID and converted into a QR code. This QR code acts as a digital identity for the product.

6. Buyer Verification
The buyer scans the QR code using a mobile device. The system extracts the Batch ID from the URL and initiates the verification process.

7. Search and Verification
The system first checks the blockchain:

- If data is found → Verified
- If not → checks cloud storage
- If still not found → displays error

8. Result Display
The system displays complete product details along with verification status, ensuring transparency and trust.



V. Modules identified:

The AgroLedger system is divided into the following functional modules:

1. Farmer Module
Allows farmers to input crop details and register product batches.

2. Storage Module
Handles data storage using:

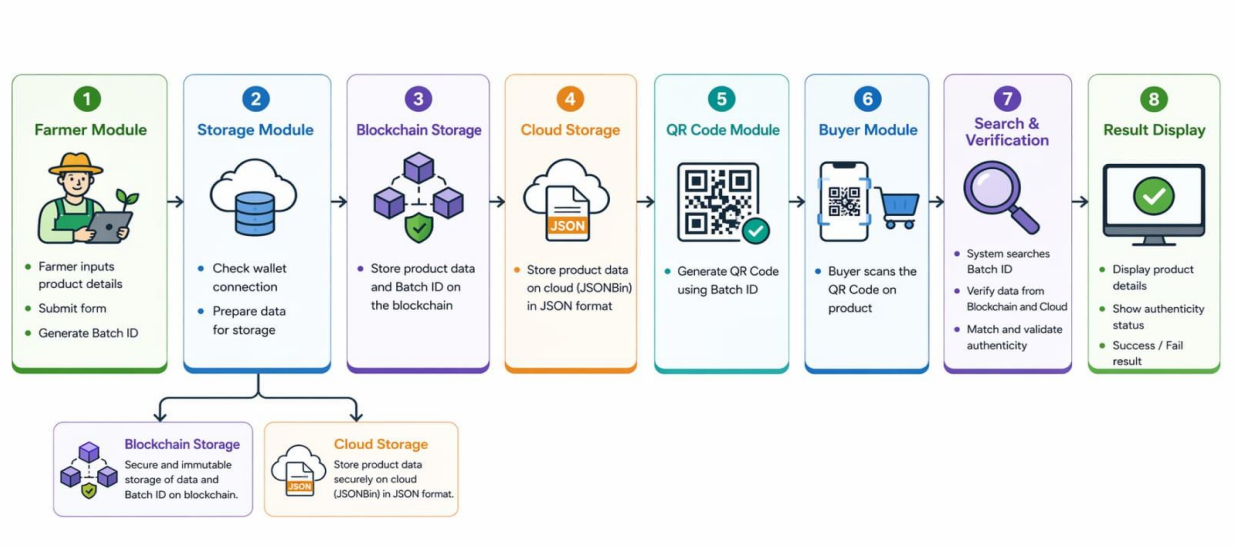
- Blockchain (secure storage)
- Cloud (fallback storage)

3. QR Code Generation Module
Generates QR codes linked to product data for easy access.

4. Buyer Module
Enables buyers to scan QR codes and access product information.

5. Search & Verification Module
Implements logic to retrieve data from blockchain or cloud and verify authenticity.

6. Result Display Module
Displays product details and verification status to the user.



VI. Work progress / plan & Implementation

Work Completed:

- Problem identification and research
- System design and architecture planning
- Frontend development using React
- Blockchain integration for secure storage
- Cloud storage (JSONBin) implementation
- QR code generation and scanning functionality
- Search and verification logic implementation
- Cross-device testing and debugging

VII. Conclusions

AgroLedger presents a practical and efficient solution to address the lack of transparency in agricultural supply chains. By combining blockchain and cloud storage, the system ensures both secure and accessible data management. Blockchain provides immutability and trust, while cloud storage guarantees smooth performance and cross-device availability.

The use of QR codes makes the verification process simple and user-friendly, allowing buyers to instantly access and validate product details. This improves confidence in product authenticity and helps in making informed purchasing decisions.

From the farmer's perspective, AgroLedger provides a digital identity for their produce, enabling them to showcase authenticity and quality. This reduces dependency on intermediaries and supports fair pricing. Overall, AgroLedger is a scalable, reliable, and user-friendly system with strong potential for real-world implementation. It contributes to building a more transparent, trustworthy, and efficient agricultural ecosystem.

IX. References:

[1] Tian, F. (2017). *A Supply Chain Traceability System for Food Safety Based on Blockchain Technology*. IEEE International Conference.

[2] Saberi, S., Kouhizadeh, M., Sarkis, J., & Shen, L. (2019). *Blockchain Technology and Its Relationships to Sustainable Supply Chain Management*. International Journal of Production Research.

[3] Casino, F., Dasaklis, T. K., & Patsakis, C. (2019). *A Systematic Literature Review of Blockchain-Based Applications: Current Status and Future Directions*. Telematics and Informatics.

[4] Kamilaris, A., Fonts, A., & Prenafeta-Boldú, F. X. (2019). *The Rise of Blockchain Technology in Agriculture and Food Supply Chains*. Trends in Food Science & Technology.

[5] Lin, J., Shen, Z., Zhang, A., & Chai, Y. (2020). *Blockchain and IoT Based Food Traceability for Smart Agriculture*. Future Generation Computer Systems.

[6] Web3.js Documentation

[7] JSONBin API Documentation